



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PART TEST
05-11-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

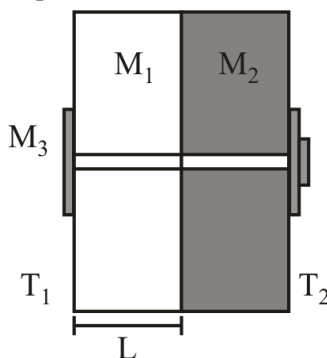
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

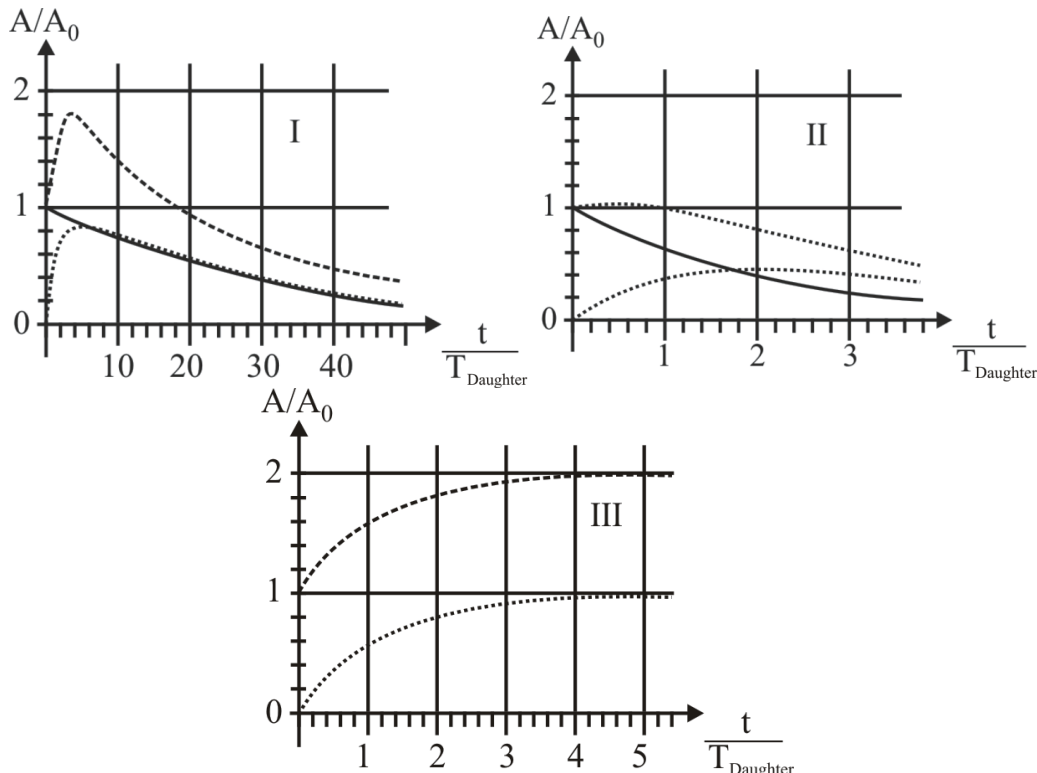
- The binding energy per nucleon of ^{10}X is 8 MeV and that of ^{11}X is 7.5 MeV where X represents an element. The minimum energy required to remove a neutron from ^{11}X is
(A) 7.5 MeV (B) 2.5 MeV (C) 8 MeV (D) 0.5 MeV
- Two different materials M_1 , M_2 of equal dimensions (length L and uniform cross-sectional area A) having thermal conductivities k_1 and k_2 respectively are connected together through a bolt M_3 . The bolt (length = $2L$, area = A/n) is of another material with thermal conductivity k_3 such that $n \gg 1$. The left, sides of the material M_1 and M_3 are kept at the temperature T_1 whereas the right sides of the material M_2 and M_3 are kept at the temperature T_2 as shown in the figure below such that $T_1 > T_2$. Assume that the bolt (M_3) is thermally insulated from materials M_1 and M_2 . The equivalent thermal resistance of this system is



- (A) $\frac{L(k_1 + k_2)}{Ak_1k_2}$ (B) $\frac{L}{(k_1 + k_2)A}$
- (C) $\frac{L(k_3 + 2n(k_1 + k_2))}{A(k_1 + k_2)nk_3}$ (D) $\frac{2nL(k_1 + k_2)}{A(2nk_1k_2 + k_2k_3 + k_3k_1)}$

Space for Rough Work

3. Three radioactive preparations are considered below. Initially, at time $t = 0$, they consist of 100% of a single radioactive isotope, the respective parent nuclide. The initial activity of the preparations is designated A_0 in each case. The direct products of decay, the daughter nuclides, are also radioactive again and decay. Further subsequent decays are no longer considered. The mother and daughter nuclides of the three preparations are:



Preparation 1 : ^{226}Ra ($T_{\text{mother}} = 1600 \text{ y}$) $\rightarrow$ ^{222}Rn ($T_{\text{daughter}} = 3.8 \text{ d}$)

Preparation 2 : ^{211}Pb ($T_{\text{mother}} = 36.1 \text{ min}$) $\rightarrow$ ^{211}Bi ($T_{\text{daughter}} = 2.14 \text{ min}$)

Preparation 3 : ^{214}Pb ($T_{\text{mother}} = 26.8 \text{ min}$) $\rightarrow$ ^{214}Bi ($T_{\text{daughter}} = 19.9 \text{ min}$)

With T_{mother} or T_{daughter} the half-lives of the respective nuclides are indicated. The graphs show the time courses of the activities A of both the mother nuclide and of the daughter nuclide and the total activity for the three preparations.

Which of the three nuclide pairs belongs to which diagram?

- (A) $\frac{\textcircled{1} \quad \textcircled{2} \quad \textcircled{3}}{\text{I} \quad \text{II} \quad \text{III}}$
- (C) $\frac{\textcircled{1} \quad \textcircled{2} \quad \textcircled{3}}{\text{III} \quad \text{II} \quad \text{I}}$

- (B) $\frac{\textcircled{1} \quad \textcircled{2} \quad \textcircled{3}}{\text{II} \quad \text{III} \quad \text{I}}$
- (D) $\frac{\textcircled{1} \quad \textcircled{2} \quad \textcircled{3}}{\text{III} \quad \text{I} \quad \text{II}}$

Space for Rough Work

4. When we press the nozzle of a deodorant, we feel cooling effect. This is due to
- (i) The liquid is cooled at the factory and is distributed to us in adiabatic containers, so the liquid remains cold.
 - (ii) The liquid coming out as small droplets so has more surface area, hence its internal energy is utilized to create the surface energy.
 - (iii) The liquid comes out through a tube which is near the bottom of the container. The colder liquid is at the base due to higher density.
 - (iv) The liquid coming out as small droplets so has more surface area, hence it can absorb heat quickly from our body.
- (A) Only i and ii are correct
- (B) Only ii and iv are correct
- (C) Only i, ii and iii are correct
- (D) Only ii, iii and iv are correct

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. In a closed cylinder of inner radius B and inner height h there is a liquid that fills a fraction k of the cylinder's inner volume. Initially the cylinder is at room temperature T_1 . The cylinder is put into a freezer with a constant temperature T_2 , which is below the melting temperature of the substance in the cylinder. It is known that the density of the substance in liquid form is ρ_0 and in solid form $\lambda\rho_0$. Neglect the heat capacity of the cylinder and its thermal expansion or contraction. The specific heat capacity of the liquid is S and the specific heat capacity in solid form is $S/2$. The heat removed from the substance in liquid form is equal to the heat removed from the substance in the solid form and also equal to the heat removed to solidify the substance completely.

(A) The latent heat of fusion is $S(T_1 - T_2)/4$

(B) The melting point is $(2T_1 + T_2)/3$

(C) The fraction of cylinder filled by the liquid finally is $k\lambda$.

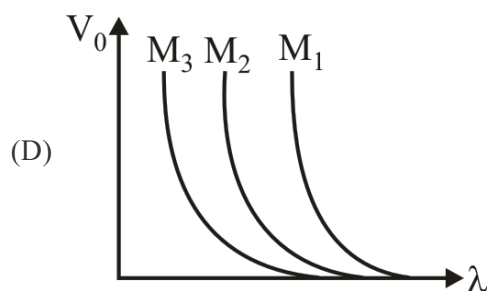
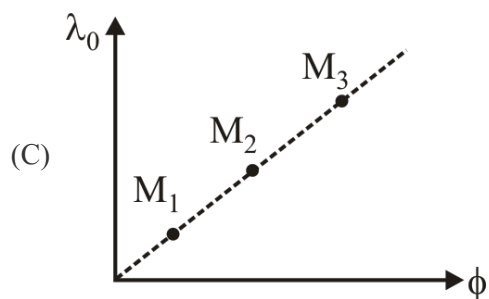
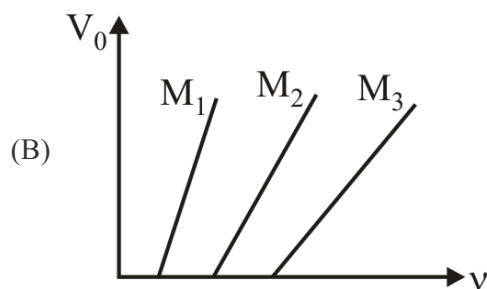
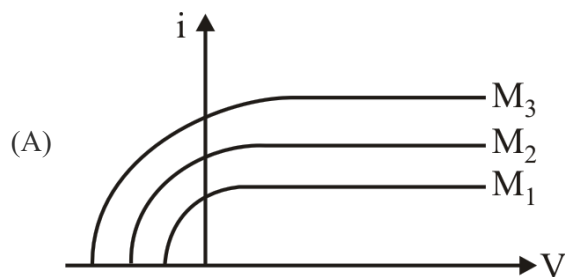
(D) The ratio of final pressure and initial pressure of air above is $\frac{T_2\lambda(1-k)}{T_1(\lambda-k)}$

Space for Rough Work

6. In an X ray tube, we gradually decrease the potential difference applied. When the minimum wavelength of emitted x rays increases to 0.25 nm, we observe that characteristic x ray spectra corresponding to K series disappears. When the wavelength increases to 1 nm, the characteristic x ray spectra corresponding to L series also disappears. Which of the following statements are correct? Assume Moseley's law, $R = 10^7/\text{m}$.
- (A) Wavelength of $K\alpha$ line is 0.33 nm
- (B) The atomic number of the target atoms is 21
- (C) Wavelength of $K\alpha$ line is 0.25 nm
- (D) Wavelength of the incident electrons in the X ray tube is being gradually increased
7. The electron in the Li^{2+} ion is replaced by a meson which has the same charge as the electron. Such an ion is called a Li^{2+} mesic ion. Let E_B be the binding energy in ground state of the hydrogen atom, and r_B the Bohr radius. You may take the mass of a meson to be 200 times the mass of electron. Which of the following statements is/are true for the Li^{2+} mesic ion? Neglect nuclear motion.
- (A) The binding energy in ground state is $1800E_B$.
- (B) The radius of the meson in the ground state is $r_B/1800$.
- (C) The angular momentum in the ground state is $200h/2\pi$, where h is Planck's constant.
- (D) The Balmer series will be in the infrared wavelengths.

Space for Rough Work

8. In a photoelectric setup, three metals M_1 , M_2 and M_3 with work functions ϕ_1 , ϕ_2 and ϕ_3 respectively ($0 < \phi_1 < \phi_2 < \phi_3$), are illuminated with light of frequency ν and wavelength λ . The currents i and voltages V are registered. The stopping potential and cutoff wavelength are denoted by V_0 and λ_0 , respectively. Which plot below could be a possible outcome of the experiment?



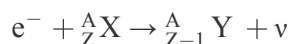
Space for Rough Work

9. Consider positron (e^+) emission from a nucleus at rest:



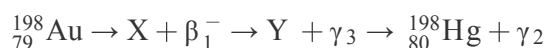
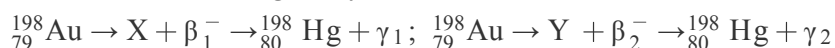
Which of the following statements is/are **CORRECT**?

- (A) The emitted e^+ has a continuous energy spectrum since the energy released in the decay is shared by e^+ and ν .
- (B) If there was no additional particle ν emitted in the decay, the energy of the emitted e^+ would be always equal to $(M_X - M_Y)c^2$, if the recoil energy of Y is neglected. (M_X and M_Y are rest masses of the nuclei X and Y, respectively, and c is the speed of light in vacuum.)
- (C) The emitted neutrino (ν) always has non-relativistic speed (i.e., speed much less than c) due to its very small rest mass
- (D) Consider the competing process to the positron emission, in which an electron (e^-) from an inner orbit of the atom (K shell) is captured by the nucleus and a neutrino is emitted.



This process is energetically allowed if e^+ emission process is energetically allowed. ("Energetically allowed" means "allowed by the principle of conservation of energy and linear momentum".)

10. Consider the following decay schemes :



The energies corresponding to X and Y with respect to ${}_{80}^{198}\text{Hg}$ are 1.088 MeV and 0.412 MeV respectively [Note the energy of ${}_{80}^{198}\text{Hg}$ is taken to be 0.0 MeV]. The atomic masses of ${}_{79}^{198}\text{Au}$ and ${}_{80}^{198}\text{Hg}$ are 197.968 u and 197.966 u, respectively, where 1 u should be taken as 931 MeV/ c^2 . Then :

- (A) the energies corresponding to γ_1 and γ_2 are 1.088 MeV and 0.412 MeV respectively.
- (B) the energies corresponding to γ_1 and γ_3 are 1.088 MeV and 0.412 MeV respectively.
- (C) the maximum kinetic energies of β_1^- and β_2^- are about 0.77 MeV and 1.45 MeV respectively.
- (D) the energy corresponding to ${}_{79}^{198}\text{Au}$ is 1.50 MeV.

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

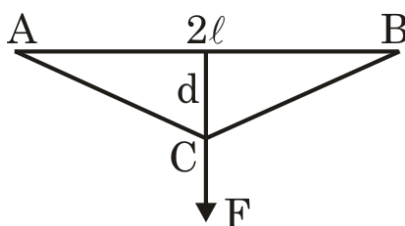
Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

-
1. A commercial food freezer maintains a temperature of -13°C . The ambient air temperature is 26°C . If heat leaks into the freezer at the continuous rate of 20 kJ/s what is the least power (in kW) necessary to pump this heat out continuously?
 2. What is the electric field at the surface of ${}^8_4\text{Be}$ nucleus? Assume $R_0 = 1.2\text{ fm}$ and charge to be uniformly distributed within the volume. If $E = 10^x\text{ N/C}$, fill $\frac{x-1}{4}$.
 3. An excited hydrogen atom at rest emits a photon and recoils with a velocity of $7/36\text{ m/s}$. $R = 10^7/\text{m}$ and $h = 6.4 \times 10^{-34}\text{ kgm}^2/\text{s}$. What is the quantum state of the electron from where photon originated?
-

Space for Rough Work

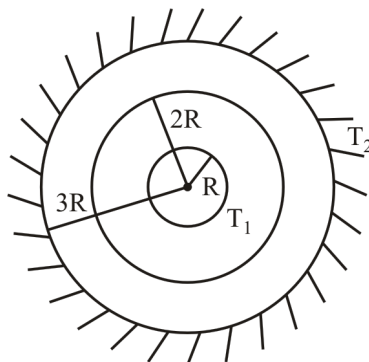
4. A tall cylindrical compressed gas tank of radius $r = 0.2\text{m}$ is made of steel that tolerates force per unit area up to $\sigma = 250\text{MPa}$. Find the largest pressure (in Mpa) of gas in the gas tank that the tank can endure. The width of the tank's wall is $t = 4\text{mm}$.
5. The air temperature in a sauna's steam room of volume $V = 10\text{ m}^3$ is $t = 90^\circ\text{C}$. Water with a mass $m = 300\text{ g}$ is cast on the stove for steam and it vaporizes immediately. Let us think hypothetically that the steam room is air tight. With what force (in kN) should the sauna visitors pull the handle of a door of area $A = 1.0\text{m}^2$ so that the door would open? The gas constant is $R = 8.3\text{ J}/(\text{mol}\cdot\text{K})$ and the molar mass of water is $\mu = 18\text{ g/mol}$. Round off the nearest integer.
6. A steel wire of area 1mm^2 is stretched without tension along a straight line with its ends fixed at A and B (figure). The wire is pulled into the shape ACB. Assume that d is very small compared to length of the wire. Young's modulus of steel is $Y = 2 \times 10^{11}\text{ N/m}^2$. Determine the pulling force F (in N)? $d = 1\text{ cm}$ and $AB = 2\text{ m}$. Fill 10F in OMR Sheet.



Space for Rough Work

7. Three concentric thin spherical shells are shown in figure. Outer most sphere can't radiate in outer space. The inner most and the outer most shells are maintained at T_1 K & T_2 K respectively. Assume the three shells behaves as black body. The steady state temperature of the middle shell is $\left(\frac{T_1^4}{x} + \frac{T_2^4}{y} \right)^{1/4}$.

Value of $\frac{x}{y}$ is



8. Two cylinders of equal masses, one made of material A and the other of material B, are heated to 50°C and placed on two large blocks of ice at 0°C . If both the cylinders have the same height, find the ratio $\left(\frac{h_A}{h_B} \right)$ on their maximum depths of penetration in the ice. Assume that no heat is lost to the surrounding and change in gravitational potential energy is not considered.

$$S_A = 0.2 \text{ cal g}^{-1}(\text{C}^\circ)^{-1}; \rho_A = 4 \text{ g cm}^{-3}$$

$$S_B = 0.1 \text{ cal g}^{-1}(\text{C}^\circ)^{-1}; \rho_B = 2 \text{ g cm}^{-3}$$

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. $(?) + \text{Br}_2 \xrightarrow[\text{sealed tube}]{\text{heated in}}$ **Metaphosphoric acid**

What is missing reactant?

(A) H_3PO_4

(B) $\text{H}_4\text{P}_2\text{O}_7$

(C) H_3PO_3

(D) Red phosphorus

2. The reaction of NiBr_2 with two equivalents of PPh_3 in CS_2 at -78°C gives a red coloured diamagnetic complex compound $[\text{NiBr}_2(\text{PPh}_3)_2]^0$. This transforms to a green coloured paramagnetic complex with the same molecular formula at 25°C . The geometry, number of unpaired electron(s) and geometrical isomer(s) in the green coloured complex, respectively are

(A) Tetrahedral, two and two

(B) Tetrahedral, two and zero

(C) Square planar, zero and two

(D) Square planar, two and two

Space for Rough Work

3. Select correct about P–O bond length in P_4O_6 & P_4O_{10}

(I) (Specified length)		(II) Bond length in pm	
(I)	P–O length in P_4O_6	(P)	141 (pm)
(II)	Endocyclic P–O length in P_4O_{10}	(Q)	165 (pm)
(III)	Exocyclic P–O length in P_4O_{10}	(R)	158 (pm)

- (A) I = P ; II = Q ; III = R
 (B) I = Q ; II = R ; III = P
 (C) I = R ; II = Q ; III = P
 (D) I = Q ; II = P ; III = R
4. Which monomeric molecule is difficult to isolate?

- (A) NO_2 (B) $(CH_3)_2Si = O$
 (C) $(CH_3)_2C = O$ (D) HF

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which option(s) is/are **INCORRECT** matching?

(A) $\text{AsH}_3 \Rightarrow$ p-s overlapping between As & H ; Triangular pyramidal molecule

(B) $\text{PH}_5 \Rightarrow \text{sp}^3\text{d}$; Triangular bipyramidal molecule

(C) $\text{GeCl}_6^{2-} \Rightarrow \text{d}^2\text{sp}^3$, octahedral ion

(D) $\text{CF}_6^{2-} \Rightarrow \text{sp}^3\text{d}^2$ octahedral ion

Space for Rough Work

6. Which is correct notation for specified molecular orbital of dioxygen.
[without using (*) notation for antibonding orbital]
- (A) H.O.M.O. = $\pi_{(g)}$
- (B) L.U.M.O. = $\sigma_{(u)}$
- (C) Molecular orbital formed by 2s-2s and has one nodal plane = $\sigma_{(g)}$
- (D) Molecular orbital which has shape  = $\sigma_{(g)}$
7. Select correct about borax.
- (A) Prepared by reaction of colemanite with Na_2CO_3
- (B) It dissolves in water to give acidic solution
- (C) It contains tetranuclear unit which has four boron, five bridging oxide & four terminal hydroxides
- (D) It produce green coloured bead $\text{Cr}(\text{BO}_2)_3$ with Cr_2O_3 on heating on Bunsen burner oxidising flame & observed in cold
8. Select the correct matching of member(s) of group 14 that show specified property.
- (A) Form most acidic oxide = Carbon
- (B) Commonly found in +2 oxidation state = Lead
- (C) Used as semiconductor = Silicon & Germanium
- (D) Tetraiodide is non existing = Lead

Space for Rough Work

9. Which compound can convert iodide ion to iodine in aqueous solution?
- (A) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (B) Cl_2 water
(C) $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ (D) $\text{H}_2\text{S}_2\text{O}_8$
10. A pale blue diamagnetic gas (Y) is best prepared by flowing paramagnetic gas (X) at 1 atm and 25°C through concentric metallized glass tubes to which low frequency power at 50-500 Hz and 10-20 kV is applied to maintain a silent electric discharge.
Gas (X) is colourless in gas form but liquid is pale blue.
Gas (Y) is used for sterilization of foodstuffs and disinfection, and also form during the generation of photochemical smog.
(X) & (Y) are the different form of a element which is about 47% by mass of earth crust.
Identify **CORRECT** statement(s) regarding (X) & (Y)
- (A) Concentration of 'Y' in the mixture of X & Y is determined by iodometrically using H_3BO_3 & $\text{Na}[\text{B}(\text{OH})_4]$ buffer.
(B) Conversion of (X) to (Y) is an exothermic process
(C) 'Y' can oxidise black lead sulphide
(D) Y is used to convert potassium manganate to potassium permanganate
-

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

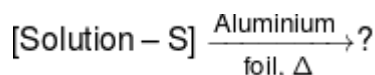
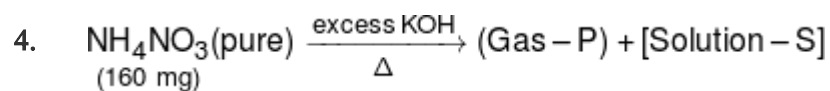
Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

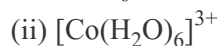
1. Consider the electrolysis procedure for purifying copper.
Suppose that a sample of copper contain the following impurities : Fe, Ag, Zn, Au, Co, Pt, Ca and Pb
The metals will be oxidised and dissolve in solution are = x
and
The metal will be unaffected and simply form the sludge that accumulate at the bottom of the cell = y
Find difference of x & y.
2. Find difference of the stereo isomers of $[Ma_3b_2c]$ & $[Ma_2b_2c_2]$
3. Borax is $Na_2B_4O_7 \cdot xH_2O$ and Kernite is $Na_2B_4O_7 \cdot yH_2O$
What is the value of (x – y)

Space for Rough Work



Number of the milimoles of Gas-P obtained in step-2.

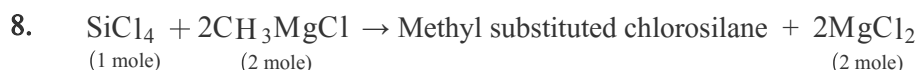
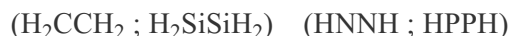
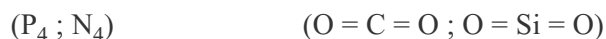
5. Find sum of the unpaired electrons in



(Given atomic number of Fe = 26, Ru = 44)

6. Write sum of optically inactive isomers of complex X, Y and Z, which are formed by reaction of ethylenediamine & $\text{Ni}_{\text{aq}}^{2+}$ in molar ratio of en : Ni is 1 : 1, 2 : 1, 3 : 1 respectively.

7. Select number of pair(s) in which 1st species is more stable as compared to 2nd



Silicone containing product of above reaction produce a compound (X) on control hydrolysis followed by condensation. (X) have same number of electrons as in discrete $\text{Si}_4\text{O}_{12}^{8-}$ ion.

Find the number of methyl group in compound (X).

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6$ are roots of equation $x^7 - 1 = 0$, then the value of $\prod_{i=1}^6 \frac{\omega - \alpha_i}{\omega^2 - \alpha_i}$ is (where ω is imaginary cube root of unity)
(A) $1 + \omega$ (B) ω (C) ω^2 (D) 1
2. Area bounded by $-8z^2 - 8\bar{z}^2 + 20z\bar{z} = 288$ is (where z is a complex number)
(A) π (B) 6π (C) 12π (D) 24π
3. If $a, b, \frac{c}{2} \in \mathbb{N}$ such that $a^2 - b^2 = c^2 + 2$, then number of possible triplets (a, b, c) is equal to -
(A) 0 (B) 1 (C) 2 (D) infinite
4. If m is the least value of $f(x) = \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 4x + 29}$ occur at $x = \alpha$, then $[m] + [\alpha]$ is equal to (where $[.]$ denotes greatest integer function)
(A) 6 (B) 7 (C) 5 (D) 4

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
5. Let $z = a + ib$ (where $a, b \in \mathbb{R}$ and such that $|2z + 3i| = |z|^2$).

Identify the correct statement(s) ?

(A) $|z|_{\text{maximum}}$ is equal to 3.

(B) Number of possible integral values of $|z|$ is equal to 3

(C) If $|z|_{\text{maximum}}$ when $z = \alpha + i\beta$ ($\alpha, \beta \in \mathbb{R}$ and $i = \sqrt{-1}$) then $(\alpha^3 + \beta^3)$ is equal to 27.

(D) If $|z|_{\text{minimum}}$ when $z = x + iy$ ($x, y \in \mathbb{R}$ and $i = \sqrt{-1}$) then $(3x^2 + 2y^2)$ is equal to 2.

Space for Rough Work

6. If $z_1, z_2, z_3 \in \mathbb{C}$ and satisfying $|2z_1 - 1| = 4$, $|2z_2 - 5| = 1$ and $|z_3 - 3| = 1$, then
- (A) maximum value of $|z_1 - z_2| = \frac{9}{2}$
- (B) least value of $|z_1 + z_2 + z_3|$ is $\frac{5}{2}$
- (C) greatest value of $|z_1 + z_2 + z_3|$ is $\frac{19}{2}$
- (D) minimum value of $|z_2 - z_3| = 1$
7. If $\sqrt{12 + 2\sqrt{15} + 4\sqrt{3} + 4\sqrt{5}} = a^{1/4} + b^{1/4} + c^{1/4}$, then $a + b + c$ is divisible by (where a, b, c are relatively prime number)
- (A) 2 (B) 10 (C) 4 (D) 5
8. Consider the system of equations
- $$2x + p^2y + 6z = 8$$
- $$x + 2y + 2qz = 5$$
- $$x + y + 3z = 4$$
- (A) Given system has no solution for $p \neq \pm\sqrt{2}$ and $q = \frac{3}{2}$
- (B) Given system has no solution for $p = \pm\sqrt{2}$ and $q = \frac{3}{2}$
- (C) Given system has infinite solution for $p = \pm\sqrt{2}$.
- (D) Given system has unique solution for $p \neq \pm\sqrt{2}$ & $q \neq \frac{3}{2}$

Space for Rough Work

9. Two tangents PA and PB are drawn to circle $S = 0$ where $P(4,0)$. If line joining to point P and centre Q cuts the circle at $C(2,0)$ and line AB at $\left(\frac{4}{3}, 0\right)$ then -
- (A) radius of circle is 1 unit
- (B) radius of circle is 2 units
- (C) $\frac{\text{area of } \triangle PAB}{\text{area of } \triangle AQB} = 8$
- (D) $PA = 2\sqrt{2}$
10. Consider $\triangle ABC$ such that AB is parallel to y-axis, BC is parallel to x-axis and centroid is at $(2, 1)$. If median through A is $y = x - 1$, then which of the following is/are true -
- (A) Equation of median through C is $x - 4y + 2 = 0$
- (B) Equation of median through C is $x + 4y = 5$
- (C) Acute angle between median through A and C is $\tan^{-1}\left(\frac{3}{5}\right)$
- (D) If length of $BC = 2$, then coordinate of D is $\left(2 - \frac{\sqrt{2}}{3}, 1 - \frac{\sqrt{2}}{3}\right)$

Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. The region bounded by the rays $\arg(z - (1 + i)) = 0$, $\arg(z - (2 + i)) = \frac{\pi}{2}$; $\arg(z - (2 + 2i)) = \pi$ and $\arg(z - (1 + 2i)) = -\frac{\pi}{2}$ is circumscribed by a circle of area A, then value of $[A]$ is (where $[.]$ denotes greatest integer function)
2. Consider the system of equations $ax + y = b$, $bx + y = a$, $ax + by = ab$, where $a, b \in \{0, 1, 2, 3, 4, 9\}$. The number of ordered pairs (a, b) for which system is consistent is equal to
3. A regular hexagon is drawn circumscribing the circle with centre $\left(1, \frac{1}{\sqrt{3}}\right)$ and radius $2.(3, \sqrt{3})$ is one vertex and other vertices are $(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)$ and (x_5, y_5) . If $\sum_{i=1}^5 (x_i - y_i) = a - \sqrt{b}$ then $a + b$ is $(a, b \in \mathbb{N})$
4. If X is a non-zero column matrix, such that $AX = \lambda X$ where λ is a scalar and the matrix A is $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$, then sum of values of λ is

Space for Rough Work

5. If largest positive integral value(s) of 'x' such that the interval $\left[\frac{x}{10}, \frac{3x}{10}\right]$ contains exactly 3 integers is I, then value of $\frac{I}{4}$ is
6. The vertex A of $\triangle ABC$ is (6, 8) & the equation of two of its medians are $y = x$ & $2x - 5y + 14 = 0$. If orthocentre of $\triangle ABC$ is (α, β) then value of $\frac{7(\alpha + \beta)}{12}$ is equal to
7. Sum of integral value(s) of x for which $|1 - x|^{\log_2(x+1)} > |x - 1|^{\log_2(4-x)}$ is
8. If $\text{adj}A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$ and $|A| > 0$, then value of $\text{tr}(A)$ is equal to
(where $\text{adj}(A)$, $\text{tr}(A)$ and $|A|$ denotes adjoint, trace of matrix A and determinant of A respectively)

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

Space for Rough Work